SE 413/513 Software Testing

Course Introduction: SE 413/513 (Software Testing)

Communication Channels:

- ***Blackboard** messages (primary choice)*
- *Email*
- *In person/Online*

  By appointment (Booking Calendar):
  Mackinac Hall (MAK) D-2-216 (Allendale)

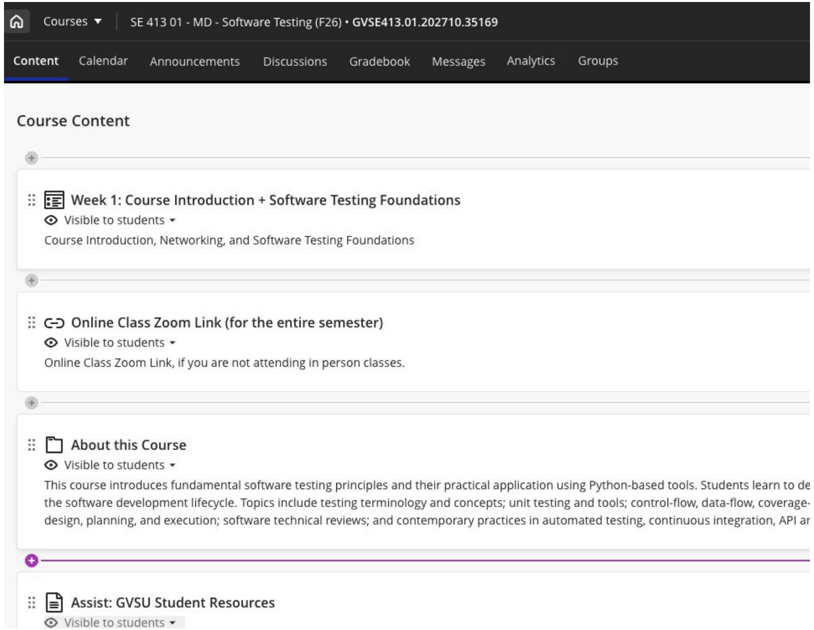
Course Faculty

Kamrul Hasan

INSTRUCTOR

Pronouns: he/him/his
Email: hasanka@gvsu.edu;
hasanka@mail.gvsu.edu

Week 1: Introduction to SE 413/513



Blackboard walk through!

Week overview

This week we will:

- Preliminary Assessment Survey (Please complete by Friday, Sept 04, 2026)
- Get to know each other (**networking**)
- Set up our **course objective, guidelines, and evaluation** procedure.
- Introduction to Software Testing: Terminology, principles, verification and validation, defects, and the testing lifecycle

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Networking!

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- **Probabilistic ML**
  - **Semi-supervised Learning**
  - **Generative AI** (example: LLMs ,ChatGPT)

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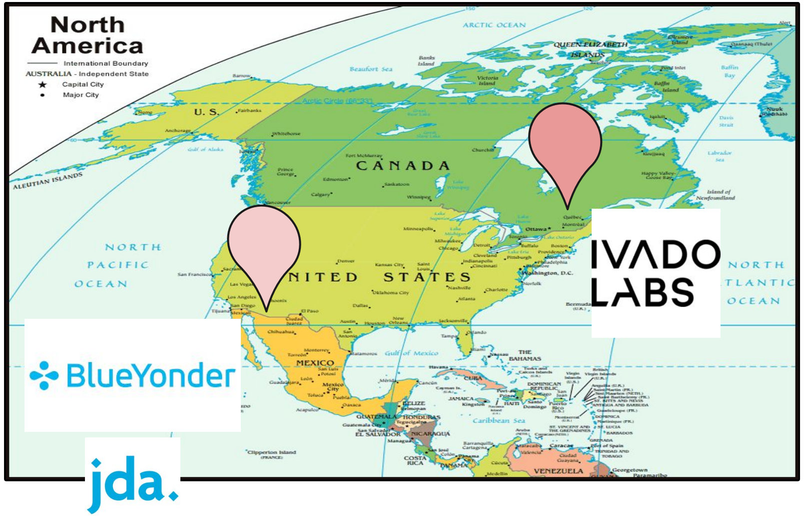

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About you!

- I hope I would get to know each of you as we progress
- We will meet in person and/or virtually as per our availability
- You will collaborate as groups
  - For discussion different topics: course specific and beyond
  - Final project
- I will seek your suggestions

Blackboard messages!

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General information (about the course)

Description:

- This course introduces students to fundamental software testing principles and their practical application using Python-based testing tools.
- Students will learn to design, implement, automate, execute, and evaluate tests across multiple levels of the software development lifecycle.
- The course integrates contemporary practices in automated testing, continuous integration, API and web testing, and selected advanced testing techniques.

Major topics include:

- Basic testing terminology and concepts
- Unit testing and unit testing tools
- Control-flow and data-flow testing
- Coverage-based and functional testing
- Integration and system testing, including test design, planning, and execution
- Acceptance testing, and
- Software technical reviews.

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Objective: Completing this course, you should be able to:

- Explain core software testing concepts, processes, and testing levels across the software development lifecycle.
- Design and implement systematic test cases using black-box, white-box, control-flow, data-flow, and coverage-based techniques, including automated unit testing.
- Execute and evaluate unit, functional, integration, system, and acceptance tests; develop test plans, interpret results, and assess code coverage.
- Apply and evaluate testing tools and practices, including version control and continuous integration, while recognizing their strengths and limitations.
- Conduct technical reviews and clearly communicate testing outcomes, including defects, risks, findings, recommendations, and testing activities.

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General information (about the course)

Textbook(s): There are no required textbooks for this course. Some good options include:

- Paul Ammann and Jeff Offutt, *Introduction to Software Testing*, 2nd ed., Cambridge University Press, 2016.
- Mauricio Aniche, *Effective Software Testing: A Developer's Guide*, Manning, 2022.
- ISTQB Certified Tester Foundation Level (CTFL) v4.0.1 Syllabus.

We will provide synthesize documentation as possible !!

Delivery Plan

General information:

- See **Blackboard** for detailed schedule
- Planning and delivery per week
- **Course materials** and **assignments** will be available on Blackboard **mostly** at the beginning of each week; sometimes as we progress.

Key dates:

- 8/31:: Class begins (weekly planning & delivery)
- 10/25-27: Spring Break
- 12/12: Class Ends
- 12/19: Semester ends

Update your Blackboard settings so you receive your emails, messages, and notifications in time.

Office: MAK D-2-216 (by appointment),
Meeting Times: Online or in person

Delivery Plan

Introduction, Unit Testing, Test Case Design, and Test Driven Development (TDD)

Week no.	Week of	Topic
1	Aug. 31	Testing Foundations
2	Sept. 7	Unit Testing
3	Sept. 14	Test Case Design
4	Sept. 21	TDD and Test Doubles
5	Sept. 28	Control-Flow Testing
6	Oct. 5	Data-Flow Testing and Coverage Concepts
7	Oct. 12	Coverage-Based Testing
8	Oct. 19	Functional Testing
9	Oct. 26	Integration Testing
10	Nov. 2	System Testing
11	Nov. 9	Acceptance Testing
12	Nov. 16	Project / Testing Workshop
13	Nov. 23	Continuous Testing
14	Nov. 30	Advanced Testing, Emerging Techniques, and AI-Assisted Testing
15	Dec. 7	Project Presentation
16	Dec. 14	Exam & Final Evaluation Week

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Delivery Plan

Introduction, Unit Testing, Test Case Design, and Test Driven Development (TDD)	Weeks 1–4
Data-Flow Testing, Coverage-Based Testing, and Functional Testing	Weeks 5–8
Integration Testing, System Testing, and Acceptance Testing	Weeks 9–11

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Introduction, Unit Testing, Test Case Design, and Test Driven Development (TDD)	Weeks 1–4
Data-Flow Testing, Coverage-Based Testing, and Functional Testing	Weeks 5–8
Integration Testing, System Testing, and Acceptance Testing	Weeks 9–11
Continuous Testing, Introduction to Advanced Testing including AI-Assisted Testing	Weeks 12–14

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Introduction, Unit Testing, Test Case Design, and Test Driven Development (TDD)

Data-Flow Testing, Coverage-Based Testing, a Functional Testing

Integration Testing, System Testing, and Acceptance Testing

Continuous Testing, Introduction to Advanc Testing including AI-Assisted Testing

Project (Learning by Doing)

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Evaluation

Grading distribution:

- Exercises /assignments/tests: 30%
- Exam / quizzes: 40%
- Final project (group + individual performance): 30%

Grade points:

A	93%	**C**	73%
A-	90%	**C-**	70%
B+	87%	**D+**	67%
B	83%	**D**	60%
B-	80%	**F**	Below 60%
C+	77%		

Note: Your final grade percentage will be rounded to the next integer percentage value. For example, an **89.1%** will round up to a **90%.**

Midterm, Quizzes, and Final Exam Policy

Written

- *Midterm exam,*
- *two quizzes,*
- *Final Exam*

will assess the concepts in machine learning covered up to that point. Additional details regarding the format and content of these assessments will be provided as the course progresses. There will be a final test. This is mainly to check student's critical, mathematical thinking, and problem solving ability as per this course is taught.

To help maintain the integrity of the course, all exams—including quizzes—will be conducted using the **Respondus LockDown Browser** with the webcam enabled. We will share detailed instructions before each exam or quiz to guide you through the process. Please take a moment to ensure that the LockDown Browser is installed and set up ahead of time. You can find helpful instructions in the guide below, and feel free to reach out to me if you have any questions or need assistance—I'm happy to help.

https://services.gvsu.edu/TDClient/60/Portal/KB/ArticleDet?ID=9865

Policy & expectation

Expectation: I expect the following to ensure your success in this course:

- check Blackboard on a regular basis for announcements, course material, and assignments
- stay up to date with required course materials.
- let me know how the class and my teaching can be improved
- adhere to the **GVSU policy of Academic Honesty** http://www.gvsu.edu/coursepolicies/

Course policy:

- Weekly reflections and homework assignments are to be completed **individually**.
- Due dates: All assignments will be due at 11:59pm Michigan time on the due date (unless otherwise stated).
- Late policy: You will lose 10% off of your maximum grade per day late, to a cap of 3 days (30% off), after which the assignment will not be accepted.

Policy & expectation

Academic Honesty:

- All students are expected to adhere to the academic honesty standards set forth by GVSU.
- In addition, students are expected to adhere to the academic guidelines as set forth by the CoC: https://www.gvsu.edu/computing/academic-honesty-30.htm
- You can learn a lot from your peers, therefore, I encourage collaboration, but passing their work off as your own is prohibited.

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With respect to all individual assignments in this course:

- Document collaboration; **no electronic transfer of code** between students is permitted.
- You are encouraged to engage in conversations in **online forums**, but do not post solutions or solicit others to complete your work for you.
- You are encouraged to talk about problems with each other in **non-technical terms** (i.e., not code).
- Ultimately, **you are responsible** for all aspects of your submissions. You should be able to explain and defend if the work is entirely your own.

Policy & expectation

Use of Generative AI Tools for Learning

- Students are permitted to use generative AI tools (e.g., ChatGPT) to support their learning, such as clarifying concepts, generating test cases, or assisting with code debugging.
- However, AI must not be used to complete or generate any part of the final work submitted for credit.
- All AI use must be clearly documented, and students are responsible for verifying the accuracy of AI-generated content.
- Improper or undocumented use of AI may constitute an academic integrity violation.

For detailed usage guidelines, responsibilities, and documentation requirements, please refer to the full document: **“Guidelines for Using Generative AI Tools in Learning”**, shared through **Blackboard**.

Useful resources

Blackboard: Course materials, assignments, grades, and announcements will be posted to Blackboard (https://lms.gvsu.edu/). It is your responsibility to stay informed.

Other academic resources: GVSU also provides opportunities for students to improve your **academic skills** through resources, such as:

- The writing center
- Computing Success Center
- Speech lab
- Research consultants
- Library Resources

Disability support : If you are in need of accommodations due to disability you must present a memo to me from Disability Support Resources (DSR), indicating the existence of a disability and the suggested reasonable accommodations. If you have not already done so, please contact the Disability Support Resources office (215 CON) by calling 331-2490 or email to dsrgvsu@gvsu.edu. Please note that I cannot provide accommodations based upon disability until I have received a copy of the DSR issued memo. All discussions will remain confidential. For more information, see https://www.gvsu.edu/dsr/

Computing Success Center: he Computing Success Center, located in MAK A-1-101, is a great place to get help or simply hang out. The Computing Success Center offers free drop-in tutoring for a variety of College of Computing courses throughout the week during the academic year.